Multi-Tenant Realtime Chat — Collaborative AI Architecture

Browser
WebSocket + REST

Mobile
WebSocket + REST

JWT Auth

ALB Ingress Controller
HTTP → REST | WebSocket → Django Channels

Service 1 — Django Core

Django App Pods (2–3)

- ASGI via Daphne / Uvicorn
- REST API (DRF) — Orgs, Users, Chat
- Django Channels — WebSocket Consumers
- JWT Auth (simplejwt)
- TenantMiddleware → injects org_id
- OrgScopedManager → filters by org_id

Celery Worker Pods

- Email on org approval
- Notifications on chat creatio
- Async side-effects from sign

bot message
back to chat

Django Signals

- post_save: Document → publishes to ai:events
- post_save: Org approved → Celery task
- post_save: Chat created → notify users

Redis Stream
ai:events

Service 2 — FastAPI AI Service

FastAPI App

- Async LLM streaming support
- Redis Stream consumer (ai:events)
- Posts agent response back to chat
- Internal service — not public facing

RAG Pipeline

- Doc ingestion + chunking
- Embeddings generation
- pgvector similarity search
- LangChain / LlamaIndex

Agentic Workflows

- Auto: doc upload → summarize
- On demand: user Q → RAG answer
- Proactive: detect doc type → flag risks
- LLM abstraction (OpenAI / Anthropic)

Redis

- Stream: chat:{chatID} — realtime chat
- Stream: ai:events — Django→FastAPI trigger
- Channel Layer — WebSocket fan-out across pods
- Celery Broker — async task queue

PostgreSQL + pgvector

- Organisation, User, AccessGroup
- Chat, ChatParticipant, Document
- pgvector — embeddings for RAG

(no separate vector DB needed)

Testing

- pytest-django — models, views, signals
- pytest-asyncio — async consumers
- channels.testing — WebSocket client
- factory_boy — test data

Collaborative AI Thread — Realtime Flow

User A uploads doc → Signal fires post_save → Redis Stream ai:events → FastAPI RAG pipeline → Agent posts summary to chat → User A + B see it in thread → User B asks question → RAG

Local Dev — Docker Desktop K8s

Jenkins CI/CD | Django Pod | FastAPI Pod | Redis Pod
PostgreSQL + pgvector | Celery Pod | pytest

Prod — AWS

EKS K8s cluster | ALB Ingress | Django Pod | FastAPI Pod | ElastiCache Redis
RDS PostgreSQL + pgvector | GitHub Actions CI/CD